RMIT Classification: Trusted

PROJECT: SCADA System Design

1. Project Description

The purpose of this project is to understand, analyze and make a software for an automated material handling system as shown in Figure 1. As you have gained sufficient knowledge about Automated System design from this class, it is time to put your knowledge to the test. In this project, you are provided a model of an automated material handling system in below figure. You are to understand the working principles, then you can analyze and design a complete software to operate the system that satisfies the following conditions:

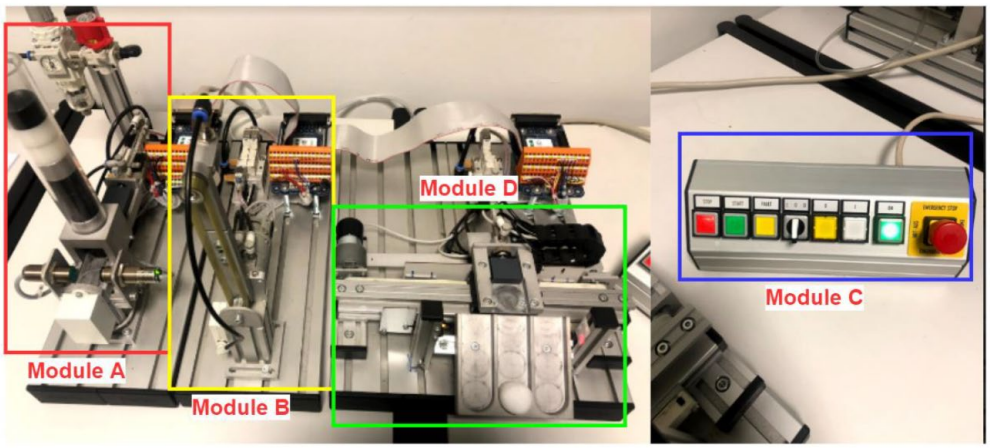


Figure 1. Model of an automated material handling system

1. There are two modes in the proposed software in which, Mode 1 (single cycle) allows it to run ONLY once time and must wait to have next commend inputs; and Mode 2 (continuous cycle) keeps the system going continuously until we put stop button OR emergency button OR if there is fault occurs (i.e., magazine empty, all containing slots full, system interfered). The start button in module C is used to initialize the process.
2. A sample of Mode 2, which can support you in understanding the working principle of the system can be seen in the Video 1 or Video 2 (attached on Canvas).
3. Three different slots in tray slider in module D will use to store workpieces such that each slot will be used to store ONLY one type of workpiece (e.g., let say slot 1 for metal workpieces; slot 2 for white plastic workpieces, and slot 3 for black plastic workpieces).
4. Time for one cycle, which from the time a workpiece is started to push by a cylinder until the time the workpiece is moved to a specific slot of the tray slider, is less than 8 seconds.
5. The automated material handling system must connect to an HMI (human-machine interface). The HMI will represent: i) the status of the system (e.g., working or stopped); ii) how many products (workpieces) are processed; iii) You also can monitor the system by the buttons on HMI (let say Start; Stop; and Emergency buttons); iv) date/month/year; and v) Your name of team name.
6. The users can select either Mode 1 (single cycle) or Mode 2 ((continuous cycle) from HMI. The user is also can monitor the system from HMI (e.g., Start; Stop; and Emergency buttons).

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Remarks:

1. Your software should be ready for demo-testing on **Thursday, September 5th, during your usual Tutorial hours**. The tests shall be conducted in the same classroom as your tutorial session. The grade for this test is worth 25% of your total grade of the project.
2. The following items are available for you to use for your project from the Robotics.
    a) HMI
    b) PCs with TIA PORTAL v19.0 + WinCC
    c) 4 PLC stations (01 S7 1500; 03 S7 1200)
    d) 10 Applications for training and study
    e) Compressed air and others.
3. Extra Support for the project and lab facilities (from TA, Mr. Tuan Huynh)
    a) To be announced soon.
    b) Location: 8.1001 (same tutorial location)

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2. Other project requirements

1. From the details of in the Project Description, perform the calculations and analysis necessary to work on the project, a **Project Report Template** (available on Canvas) will provide you for your reference. **The submission deadline for the Report is Friday, August 29, 2025, by 21:00 PM**.
2. All project team members are required to attend the **Demonstration session on Wednesday, September 3, 2025 during your usual Tutorial hours**, during your usual Tutorial hours, in the Robotics Lab, room 8.1.001. Absence from the session means a zero for the testing portion of that student.
3. The Report should base on the format of the template, which is a high level of detail with S.M.A.R.T. (Specific, Measurable, Achievable, Relevant, Time-based) descriptions should be addressed.
4. All sections within the report must be fulfilled. In case it cannot be filled, note the reason why. Make sure you discuss the reason with the Lecturer beforehand, otherwise you may lose points to that portion.
5. All submitted work, and report must be original. **If plagiarism is detected, penalties and disciplinary actions will apply per RMIT Code of Conduct**

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